DEVOPS BOOTCAMP | WEEK 01

DVP-Week01-Assignment — Linux Fundamentals

Environment: Ubuntu 24.04.4 LTS (Noble Numbat), running in Oracle VirtualBox

Evidence: Original Ubuntu terminal screenshots supplied by the student. All outputs below are interpreted from the attached evidence; sample outputs have not been substituted.

Submission date: 18 October 2026

1. Linux Distribution and Version

Commands documented

lsb_release -a
cat /etc/os-release

Explanation and observations

The commands identify the installed Linux distribution and release. Both outputs confirm Ubuntu 24.04.4 LTS, release 24.04, codename noble. The message “No LSB modules are available” is informational and does not invalidate the distribution details.

```
dinesh-a@dinesh-a-VirtualBox:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.4 LTS
Release:        24.04
Codename:       noble
dinesh-a@dinesh-a-VirtualBox:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.4 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/p
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
```

Evidence: Ubuntu distribution and version

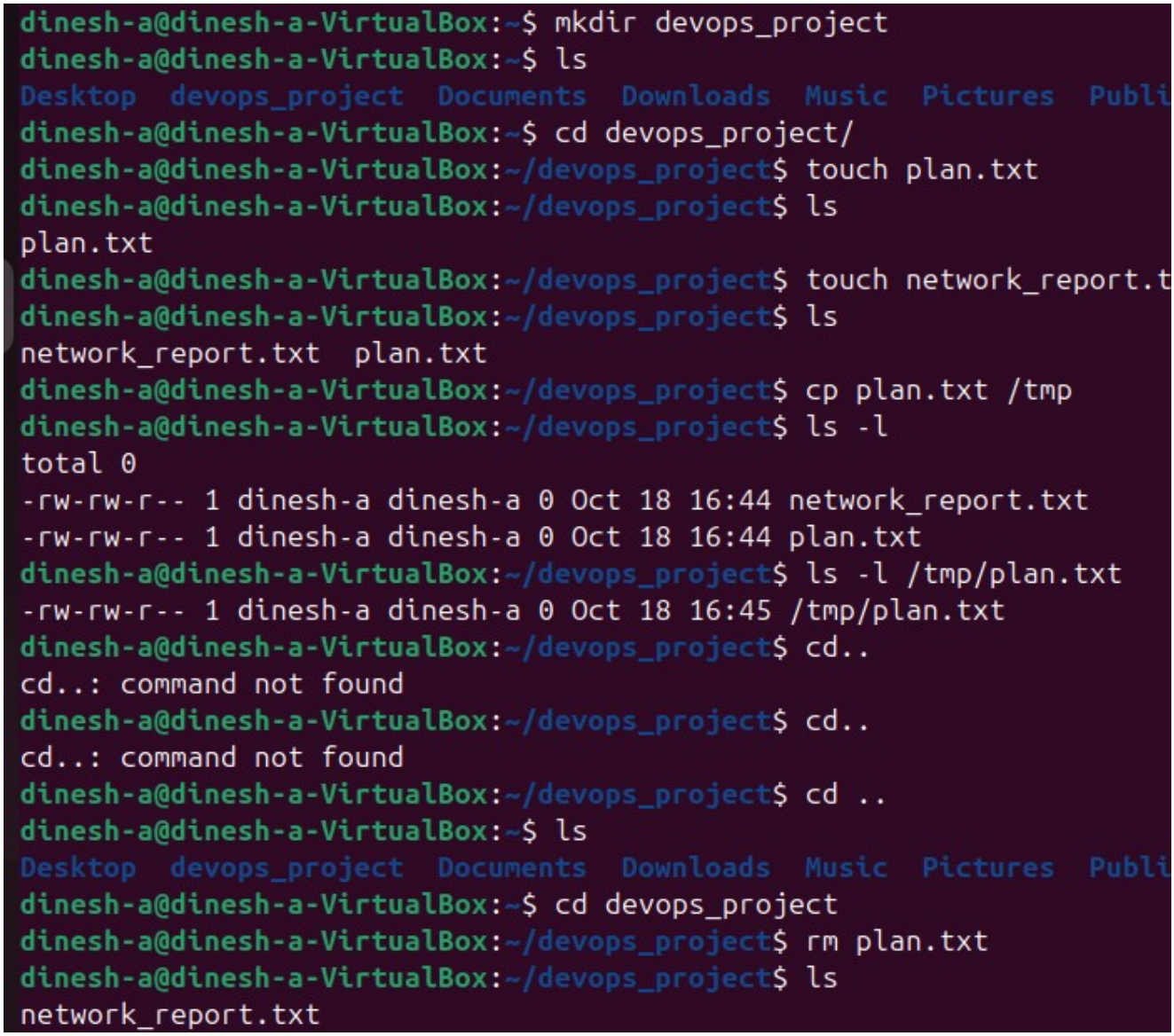
2. Files and Directories

Commands documented

```
mkdir devops_project
cd devops_project
touch plan.txt
touch network_report.txt
cp plan.txt /tmp
ls -l /tmp/plan.txt
rm plan.txt
ls
```

Explanation and observations

The screenshots show a directory named devops_project created under the user's home directory, both requested files created, plan.txt copied to /tmp, and the original removed. The final listing shows network_report.txt remains. Important path difference: the documented working directory is /home/dinesh-a/devops_project, not the explicitly requested /home/devops_project. The exact required path should be created and verified before final submission.



Evidence: File creation, copy, deletion and verification

3. Users, Groups and File Permissions

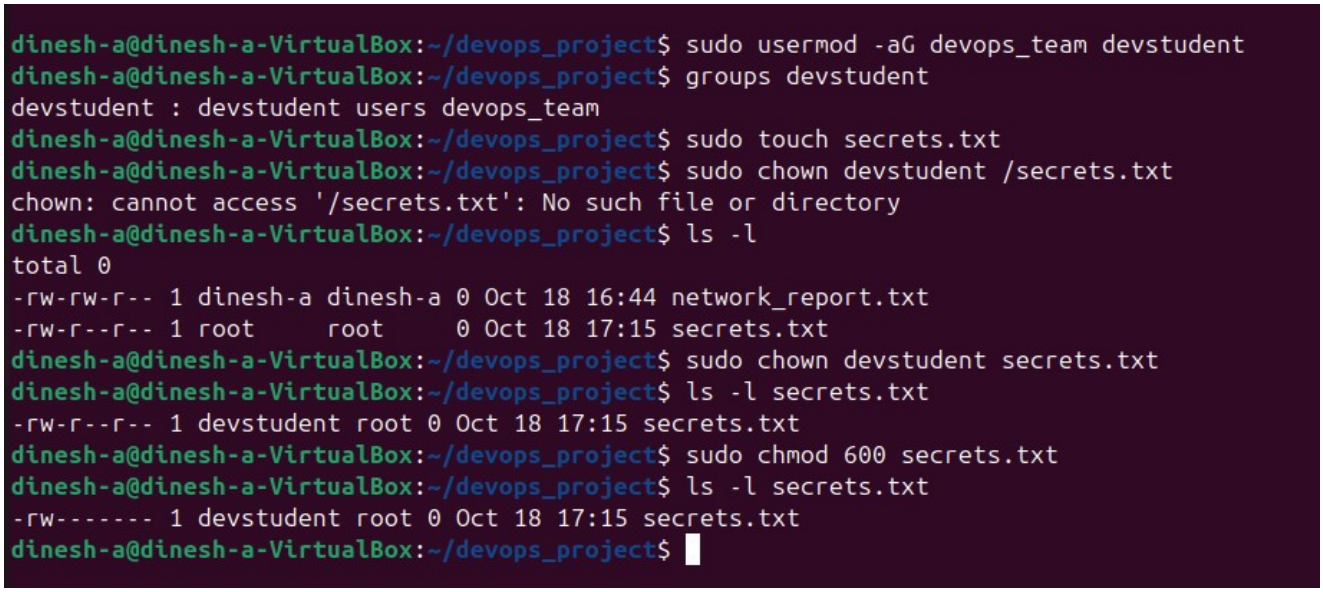
Commands documented

sudo usermod -aG devops_team devstudent
groups devstudent
sudo touch secrets.txt
sudo chown devstudent secrets.txt
sudo chmod 600 secrets.txt
ls -l secrets.txt

Explanation and observations

The groups output confirms devstudent belongs to devops_team. After an initial incorrect absolute-path chown attempt, the corrected command successfully changes ownership to devstudent. The final

permissions -rw------- represent owner read/write only (600); group and others have no access. The screenshot does not show the initial useradd and groupadd creation commands, but it confirms the user and group exist.



Evidence: Group membership, ownership and 600 permissions

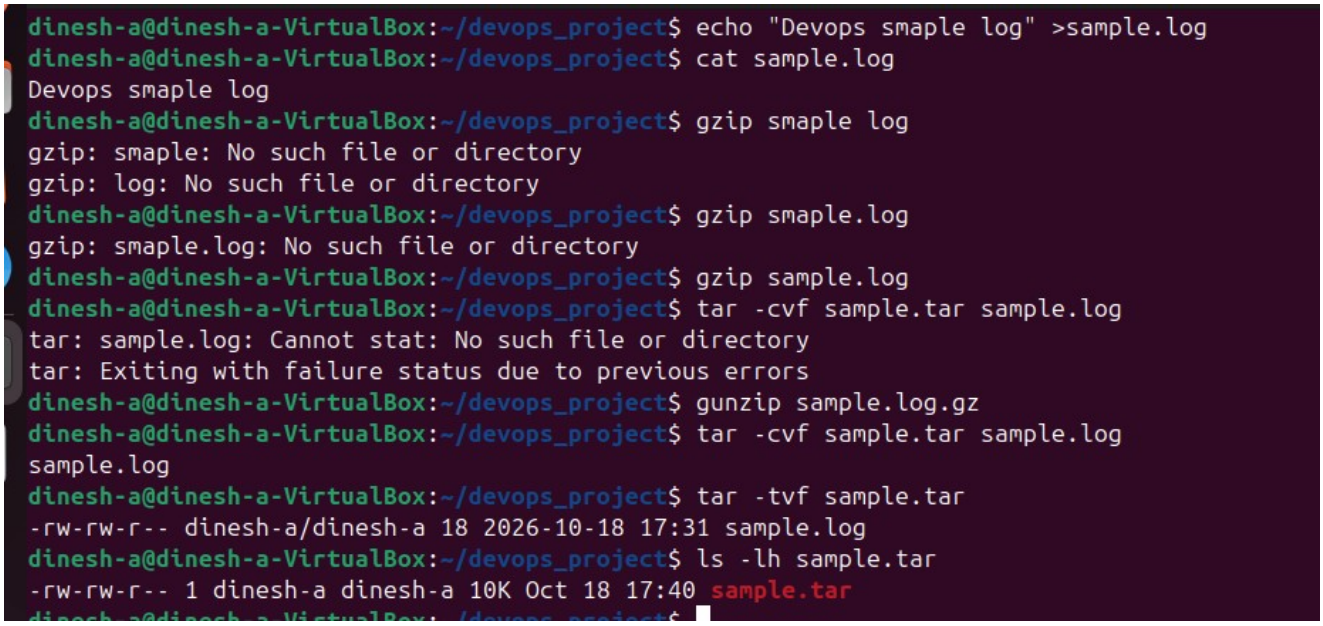
4. File Compression and Archiving

Commands documented

echo "Devops smaple log" >sample.log
cat sample.log
gzip sample.log
tar -cvf sample.tar sample.log
tar -tvf sample.tar
ls -lh sample.tar

Explanation and observations

A dummy log file was created using echo. Several attempts failed because of spelling or missing-file issues; these are visible in the evidence. A later tar -cvf command succeeds, and tar -tvf lists sample.log inside sample.tar. This verifies TAR archiving. gzip was attempted, but the screenshot does not clearly show a final .gz file; verify gzip compression separately. Note: tar -cvf creates an uncompressed TAR archive; tar -czvf creates a gzip-compressed TAR archive.



Evidence: Log file, gzip attempts, TAR archive and listing

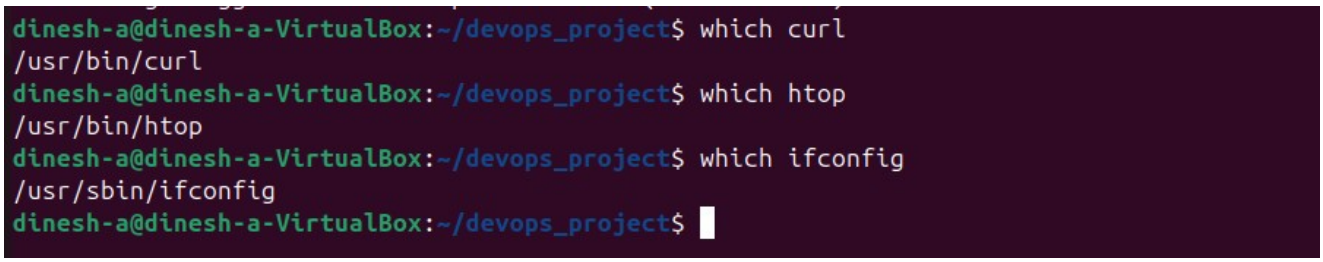
5. Package Management

Commands documented

which curl
which htop
which ifconfig
curl --version
htop --version
ifconfig --version

Explanation and observations

The which outputs locate curl at /usr/bin/curl, htop at /usr/bin/htop, and ifconfig at /usr/sbin/ifconfig. Version outputs show curl 8.5.0, htop 3.3.0, and net-tools 2.10. These outputs demonstrate that the utilities are available. The original apt installation command is not visible in the supplied screenshots, so the installation step itself is not directly documented.



Evidence: Executable paths

```
dinesh-a@dinesh-a-VirtualBox:~/devops_project$ curl --version
curl 8.5.0 (x86_64-pc-linux-gnu) libcurl/8.5.0 OpenSSL/3.0.13 zlib/1.3 brotli/1.1.0 zstd/1.5.5 libidn2/2.3.7 libpsl/0.21
.2 (+libidn2/2.3.7) libssh/0.10.6/openssl/zlib nghttp2/1.59.0 librtmp/2.3 OpenLDAP/2.6.10
Release-Date: 2023-12-06, security patched: 8.5.0-2ubuntu10.16
Protocols: dict file ftp ftps gopher gophers http https imap imaps ldap ldaps mqtt pop3 pop3s rtmp rtsp scp sftp smb smb
s smtp smtps telnet tftp
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL t
hreadsafe TLS-SRP UnixSockets zstd
dinesh-a@dinesh-a-VirtualBox:~/devops_project$ htop --version
htop 3.3.0
dinesh-a@dinesh-a-VirtualBox:~/devops_project$ ifconfig --version
net-tools 2.10
dinesh-a@dinesh-a-VirtualBox:~/devops_project$
```

Evidence: Installed utility versions

6. Networking Commands

6.1 ip a / ifconfig

ip a
ifconfig

Lists network interfaces and their addresses. The screenshot shows the loopback interface (127.0.0.1) and enp0s3 with private IPv4 address 10.0.2.15/24, consistent with VirtualBox NAT networking.

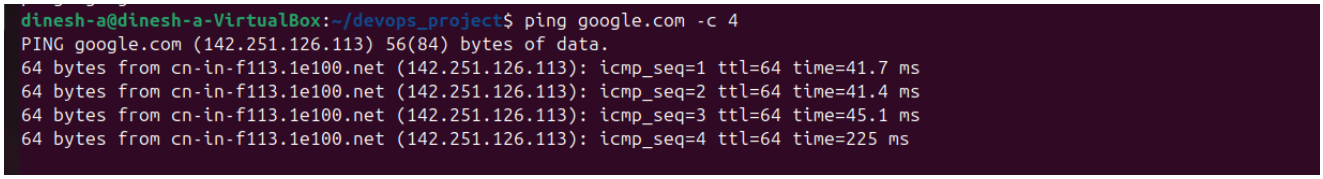
```
dinesh-a@dinesh-a-VirtualBox:~/devops_project$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:85:3c:09 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 65446sec preferred_lft 65446sec
    inet6 fd17:625c:f037:2:586a:54be:b6af:ce88/64 scope global temporary dynamic
       valid_lft 86087sec preferred_lft 14087sec
    inet6 fd17:625c:f037:2:a00:27ff:fe85:3c09/64 scope global dynamic mngtmpaddr
       valid_lft 86087sec preferred_lft 14087sec
    inet6 fe80::a00:27ff:fe85:3c09/64 scope link
       valid_lft forever preferred_lft forever
dinesh-a@dinesh-a-VirtualBox:~/devops_project$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fd17:625c:f037:2:a00:27ff:fe85:3c09  prefixlen 64  scopeid 0x0<global>
        inet6 fe80::a00:27ff:fe85:3c09  prefixlen 64  scopeid 0x20<link>
        inet6 fd17:625c:f037:2:586a:54be:b6af:ce88  prefixlen 64  scopeid 0x0<global>
        ether 08:00:27:85:3c:09  txqueuelen 1000  (Ethernet)
        RX packets 45336  bytes 47207753 (47.2 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 9714  bytes 665399 (665.3 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Evidence: Network interfaces and private IP

6.2 ping google.com -c 4

ping google.com -c 4

Sends four ICMP echo requests to test reachability and response time. The screenshot shows four replies from a Google address, with response times including 41.7 ms, 41.4 ms, 45.1 ms and 225 ms. The final packet-loss summary is not visible.

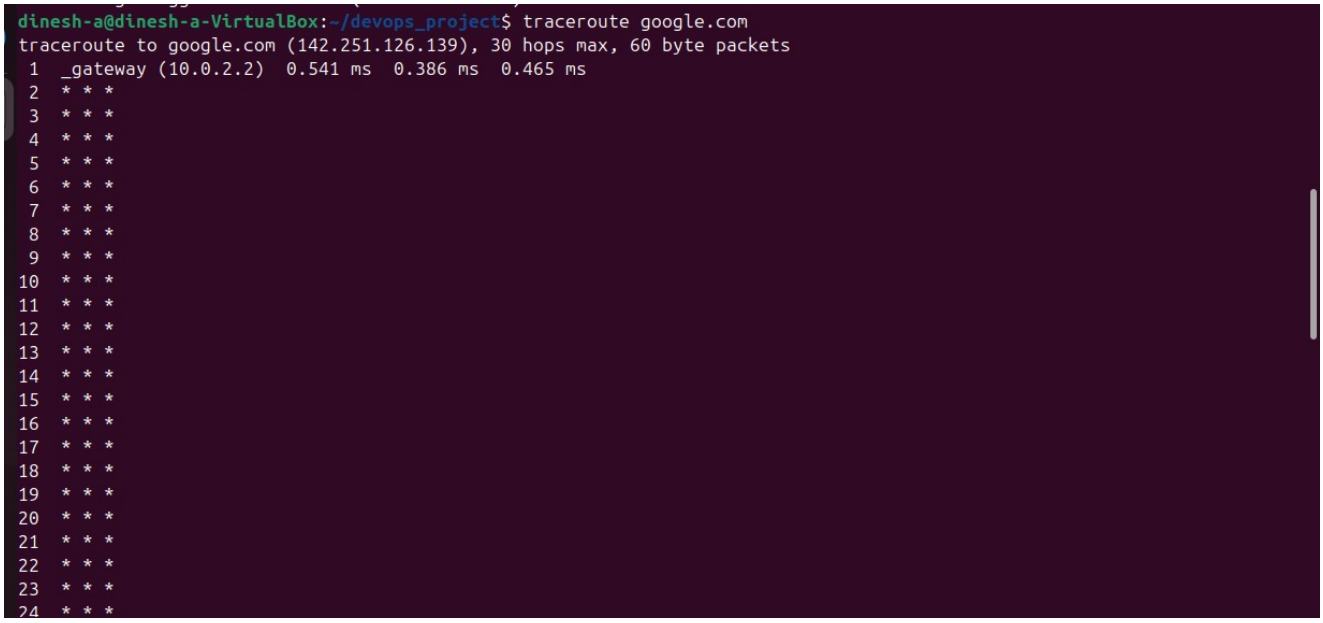


Evidence: Four ping replies

6.3 traceroute google.com

traceroute google.com

Traces the network path toward Google through successive hops. The first hop is the VirtualBox gateway 10.0.2.2. Subsequent hops show asterisks, meaning the probes received no reply before timeout; this can happen because of firewall or network filtering and does not alone prove loss of connectivity.



Evidence: Traceroute with nonresponding hops

6.4 nslookup google.com

nslookup google.com

Queries DNS to translate google.com into IP addresses. The resolver shown is 127.0.0.53, and the output returns multiple IPv4 and IPv6 addresses for Google.

```
dinesh-a@dinesh-a-VirtualBox:~/devops_project$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.126.113
Name:   google.com
Address: 142.251.126.101
Name:   google.com
Address: 142.251.126.100
Name:   google.com
Address: 142.251.126.102
Name:   google.com
Address: 142.251.126.138
Name:   google.com
Address: 142.251.126.139
Name:   google.com
Address: 2404:6800:4000:1015::64
Name:   google.com
Address: 2404:6800:4000:1015::66
Name:   google.com
Address: 2404:6800:4000:1015::8b
Name:   google.com
Address: 2404:6800:4000:1015::65
```

Evidence: DNS resolution

6.5 netstat -tuln

netstat -tuln

Displays listening TCP and UDP sockets with numeric addresses and ports. The screenshot includes local DNS-related ports such as 53 and 5353, along with other listening services.

6.6 curl ifconfig.me

curl ifconfig.me

Requests the public-facing IP address as observed by the remote website. The screenshot shows 61.2.154.120 at the time of the command; this may change and differs from the VirtualBox private IP address.

```
dinesh-a@dinesh-a-VirtualBox:~/devops_project$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tcp6       0      0 ::1:631                 :::*                    LISTEN
udp        0      0 127.0.0.54:53           0.0.0.0:*
udp        0      0 127.0.0.53:53           0.0.0.0:*
udp        0      0 0.0.0.0:37795           0.0.0.0:*
udp        0      0 0.0.0.0:5353            0.0.0.0:*
udp6       0      0 :::43056                :::*
udp6       0      0 :::5353                 :::*
dinesh-a@dinesh-a-VirtualBox:~/devops_project$ curl ifconfig.me
61.2.154.120dinesh-a@dinesh-a-VirtualBox:~/devops_project$
```

Evidence: Listening sockets and public IP lookup